Guowei Huai

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Research Interests

Dexterous manipulation and real-world robot learning, including reinforcement learning and the use of human data for robot policy pre-training and post-training. Related interests include contact-rich tool use, in-hand manipulation, teleoperation, and tactile sensing.

Education

HKUST (Guangzhou) Sept. 2026 – Present
PhD student, Robotics and Autonomous Systems Guangzhou, China
Academic Advisor: Jie Song

HKUST (Guangzhou) Sept. 2024 – Sept. 2026
MPhil, Robotics and Autonomous Systems GPA: 4.04/4.30 Guangzhou, China
Red Bird MPhil (RBM) Program
Academic Advisor: Jie Song | Project Advisor: Arthur Kar Leung Lin

Beijing Institute of Technology, Zhuhai Sept. 2019 – June 2023
B.Eng., Computer Science GPA: 89.8/100 Zhuhai, China

Publications

HiFun: A Hierarchical Framework for Efficient Functional Dexterous Manipulation Learning
Linyi Huang, **Guowei Huai**, Weibin Liu, Shulong Jiang, Ping Tan, Weixuan Zhang, Hui Zhang, Jie Song.
Conference on Robot Learning (CoRL), 2026. Accepted. Project Paper arXiv

Research Projects

HiFun: A Hierarchical Framework for Efficient Functional Dexterous Manipulation Learning
HKUST (Guangzhou), HKUST & ETH Zurich | Core contributor; second author May 2025 – May 2026

- Contributed to a hierarchical real-world RL system that separates contact-aware hand skills from arm control and skill activation for functional tool use.
- Evaluated BC, diffusion policy, human-in-the-loop RL, retargeting, and replay baselines on Franka Research 3 with XHand and Sharpa; analyzed contact failures, timing errors, and robustness under perturbations.
- The complete framework achieved 98.3% mean success across six tasks and two hands, with at most 60 minutes of online HIL training per task; the full pipeline averaged approximately 109 minutes per task.

Research Projects

Goal-Conditioned In-Hand Object Reorientation

HKUST (Guangzhou) | MPhil thesis project Dec. 2025 – Present

- Developed a goal-conditioned in-hand rotation framework for object reorientation, including elongated objects, with a desired object pose as the control goal.
- Formulated any-grasp-to-any-grasp transitions: reorient an object from an initial grasp toward a target grasp by conditioning the policy on the target object pose.
- Built a two-stage reinforcement-learning pipeline in IsaacLab, learning force-closure grasps before multi-axis object rotation with tactile feedback on Sharpa Hand.

Dexterous Teleoperation and Multimodal Data Platform

HKUST (Guangzhou) | Red Bird MPhil (RBM) team project May 2025 – Mar. 2026

- Developed optimization-based retargeting for Manus gloves and Vision Pro using vector and fingertip objectives; unified both inputs in one repository with a shared interface for Inspire Hand, Leap Hand, XHand, and Sharpa.
- Integrated GELLO arm control with Manus hand retargeting for coordinated Franka arm–hand demonstrations; also supported a 3D mouse and Vision Pro end-effector control.
- Integrated piezoresistive, electromagnetic, and vision-based tactile sensors; synchronized tactile streams, robot states, and demonstrations for real-world RL and imitation learning.

Diffusion-Based Visual Imitation Learning

HKUST (Guangzhou) course project | Contributor Code Report Mar. 2025 – May 2025

- Reproduced DP3 and RDT-1B on AIRBOT Play; processed 490 Airbot500 teleoperation episodes through workspace cropping, farthest-point sampling, and Zarr conversion.
- Analyzed real-robot evaluations reporting 72% seen-task success and 30% unseen-container generalization.

Teaching & Service

Reviewer, ICRA 2026 Workshop on Manipulation Robustness May 2026

Course instructor & designer, Datawhale May 2023 – Sept. 2024

Designed AI4S competition courses, built baselines for beginners, and led online teaching sessions.

Selected Honors

First Prize, Bionic Robot Innovation Competition Oct. 2025
HKUST (Guangzhou); MoSense team
Quarterfinalist (Top 8), 1st WBCD Competition at ICRA May 2025
Principal's First-Class Scholarship, BITZH; 1/2000 Apr. 2023
Third Prize, JD Programming and Innovative Application Competition June 2022
National First Prize, Chinese Teddy Cup Data Analysis Competition Dec. 2021
Climbing Plan Science and Technology Innovation Project Award May 2021

Technical Skills

Robot learning: reinforcement and imitation learning, diffusion policy, PyTorch, JAX.
Robotics systems: ROS, Isaac Gym/Lab, dexterous retargeting, tactile sensing, multimodal data collection.
Development: Git workflows, Claude Code, Codex.